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July 16, 2026 *By electronic transmission*

Ms. Aicha Evans
Chief Executive Officer
Zoox, Inc.
Foster City, California

Re: Exploratory inquiry — exemption-grade evidence and a physical fail-closed layer for a vehicle with no manual controls

Dear Ms. Evans:

I am the founder of Ghost Autonomy, a pre-silicon research program based in Halifax, Canada, developing what we call an **enforcement-and-accountability layer for bounded autonomy**: a risk-monotone actuation boundary terminating in non-programmable analog hardware, an evidence-admissibility discipline applied *before* sensor fusion, and a constitutionally governed memory in which every change to a system's operating truth is a logged, challengeable event. The program's distinguishing practice is evidentiary: every claim in our corpus carries a graded standing — measured, proven, proposed, or notional — and no claim is permitted to exceed it. This letter observes the same rule.

Zoox has done the two hardest things in this industry, in their hardest order: it built a genuinely novel vehicle, and then carried it into the federal system — the first-ever AVEP demonstration exemption in August 2025, public service in Las Vegas and San Francisco, and series production beginning at Hayward this month. We write because your program's next phase is, explicitly, an argument before NHTSA: the pending Part 555 commercial petition asks the agency to accept the assumptions on which a vehicle without a steering wheel or pedals is safe. For a petition of that kind, the set of assumptions on which you continue is not merely critical — it is, in effect, the filing itself. And as Hayward ramps toward the ten-thousand-units-per-year capacity you have described, every implicit assumption is multiplied by the production count.

> *At this point in your program's trajectory, we believe the most consequential engineering object you own is the set of assumptions on which it will continue — and that this set now deserves explicit registration, physical enforcement, and independent audit.*

The offer. Ghost is prepared to pivot its development to maximize compatibility with the Zoox architecture: a stateless guard ECU that presumes nothing about the vehicle above it — bidirectional operation, four-wheel steering, and the absence of manual controls included; conformance-instrument output formatted as claims-argument-evidence exhibits directly usable in exemption and California filings; and an admissibility layer parameterized to your four-pod, roughly forty-sensor suite, thermal and acoustic channels included.

What Ghost brings — specifically, and at its true standing.

1. **A measured boundary primitive.** An electrically isolated analog safety veto with ∼32 ns

intervention latency, witnessed on FPGA **[MEASURED]** — deliberately the only latency figure our corpus reports as measured rather than projected.

2. **Proven envelope mathematics.** The risk-monotone actuation algebra in its box case: projection is coordinatewise clipping; enforcement is firmly non-expansive (1-Lipschitz); the two-stage clamp is exact in the feasible case **[PROVEN]**.
3. **A fully specified refusal chain.** Five stages (S0–S4), from trusted numerics through fidelity gating, layered safety filtering, and risk-monotone envelopes to the analog veto, with six acceptance criteria (G1–G6) restated as a conjunctive, fail-closed conformance gate — published as criteria, not achievements **[PROPOSED]**.
4. **An admissibility-before-fusion evidence discipline.** Sensors as witnesses under integrity, alignment, scope, and lineage gates; correlated witnesses counted once; inadmissible evidence excluded rather than averaged in — sensor-agnostic by construction **[PROPOSED]**.
5. **A constitutional memory doctrine.** Append-only canon, anti-silent-drift amendment semantics, verification burden that increases with residence time, and recall as a priced contract (Governed Knowledge Systems trilogy, July 2026) **[NOTIONAL]**.
6. **A self-auditing corpus.** An 89-page consolidated foundational corpus (May 2026); a 94-page technical documentation edition, v3.0 (June 2026); and a 367-page compendium whose provenance ledgers grade every claim — including a published coverage map (68 subcategories: 21 full, 28 high, 9 partial, 10 gap) and a formal proof-debt ledger (P-1–P-7) naming what remains unproven.
7. **Empirical provenance, honestly scoped.** Physics-informed reduced-order kernels on elliptic-integral/AGM primitives with machine-precision convergence **[PROVEN]**, validated at $R^2 \approx 0.78$–0.87 on 200+ experimental datasets within their demonstrated envelope **[MEASURED]** — in a non-automotive domain, with automotive transfer registered in our own ledger as an open validation obligation.

What this could be worth to your program — specifically.

1. **Exemption-grade artifacts.** A machine-readable claims-argument-evidence structure aligned with the quantitative safety-case framework you already publish — built to be quoted in a Part 555 record rather than summarized for it.
2. **The strongest available answer to the no-driver question.** A physically fail-closed, non-programmable refusal path with a hardware-witnessed intervention primitive — an independent layer beneath your software stack, for a vehicle that has no human fallback by design.
3. **A shorter recall cycle.** Your 2025 software recalls were handled responsibly; signed, replayable decision traces shorten exactly that investigation-to-remedy loop and document it for the agency.
4. **Conformance gating at production scale.** The conjunctive gate as a release criterion at Hayward — assumptions checked per build, not per incident.

What Ghost is not. We have no fabricated silicon, no vehicle deployment, and no fleet data; our measured hardware evidence is presently a single number. We state this plainly because we regard the statement itself as an asset — a partner adopting our layer inherits a provenance ledger, not a marketing narrative — and because it is precisely why a compatibility-first alignment now, while our interfaces are still being fixed, costs little and shapes much.

Our inquiry is deliberately modest: *is such an alignment of interest?* As a first, zero-obligation step, we would run our conformance instrument against a trace or scenario format of your choosing and return the results with their standings — an artifact your team keeps regardless of what follows. Our two strategy volumes and the published corpus are available on request; deeper documentation under mutual NDA.

Respectfully,

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Evidentiary standings used in this letter — **[MEASURED]** witnessed on hardware; **[PROVEN]** a mathematical result; **[PROPOSED]** specified and defended, not yet validated; **[NOTIONAL]** a research direction, quotable as intent only. It is our house rule that no statement may exceed its standing; we extend the same rule to this letter.